



Neuro Shot Team

A Flu Shot Learning Project: Feature Engineering & Modeling

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Goal:

- to enrich the feature space with **domain-driven, low-cardinality features** that capture behavioral patterns, vaccine attitudes, and health vulnerability while remaining compatible with the existing encodings

Result:

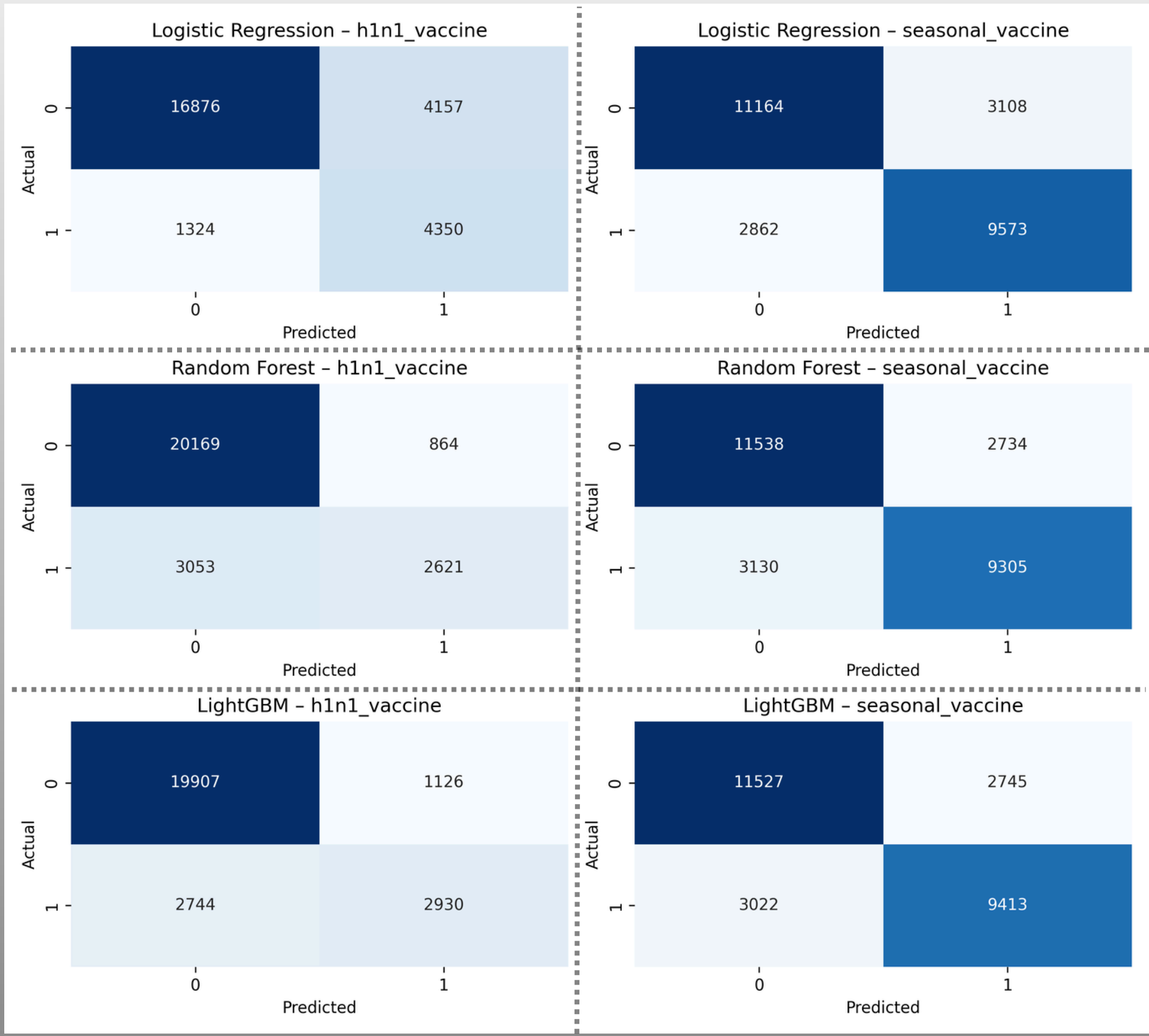
- a compact, interpretable feature set that condenses behavioral, attitudinal, socioeconomic, and medical information into actionable predictors for both modeling accuracy and real-world decision support

Feature Engineering

Feature Group	Feature Names	Purpose / What They Capture	Key Rationale
Behavioral Indices	PBI	Preventive health habits	Summarizes multiple binary behavior variables into a single scale (0–1).
Attitude Scores	PVAS_h1n1, PVAS_seas	Vaccine trust & perceived risk	Compact representation of effectiveness, risk, and side-effect opinions.
Knowledge Index	HKI	Awareness & concern	Combines concern and knowledge into a general awareness score.
Doctor Recommendation Flags	doctor_any	Exposure (or lack) to professional vaccine advice	Indicates whether a doctor recommended at least one vaccine.
Socioeconomic Flags	below_poverty, low_edu, housing_insecure, no_job, no_insurance, is_married, is_senior	Economic & demographic constraints	Capture access barriers and key sociodemographic traits.
Health Risk Index	HRI	Composite health vulnerability measure	Quantifies clinical and demographic risk factors (0–4 scale).
Interaction Features	PVASxPBI_h1n1, PVASxDoctor_seas, PBIxNoInsur, SeniorxChronic, PovertyxRent	Combined behavior–attitude or risk–access effects	Capture non-linear relationships relevant to uptake decisions.
Aggregated Target Encodings	industry_occ_te_mean	Occupation–industry vaccination tendency	Smooth summary of both target-encoded job features.

Model Selection

Confusion Matrices



Model	H1N1 Vaccine AUC	Seasonal Vaccine AUC	Macro AUC	Key Traits
LightGBM	0,865	0,860	0,863	Fast, robust, interpretable, best overall
Random Forest	0,860	0,854	0,857	Strong but less efficient and less interpretable
Logistic Regression	0,861	0,853	0,857	Simple and interpretable, slightly less flexible

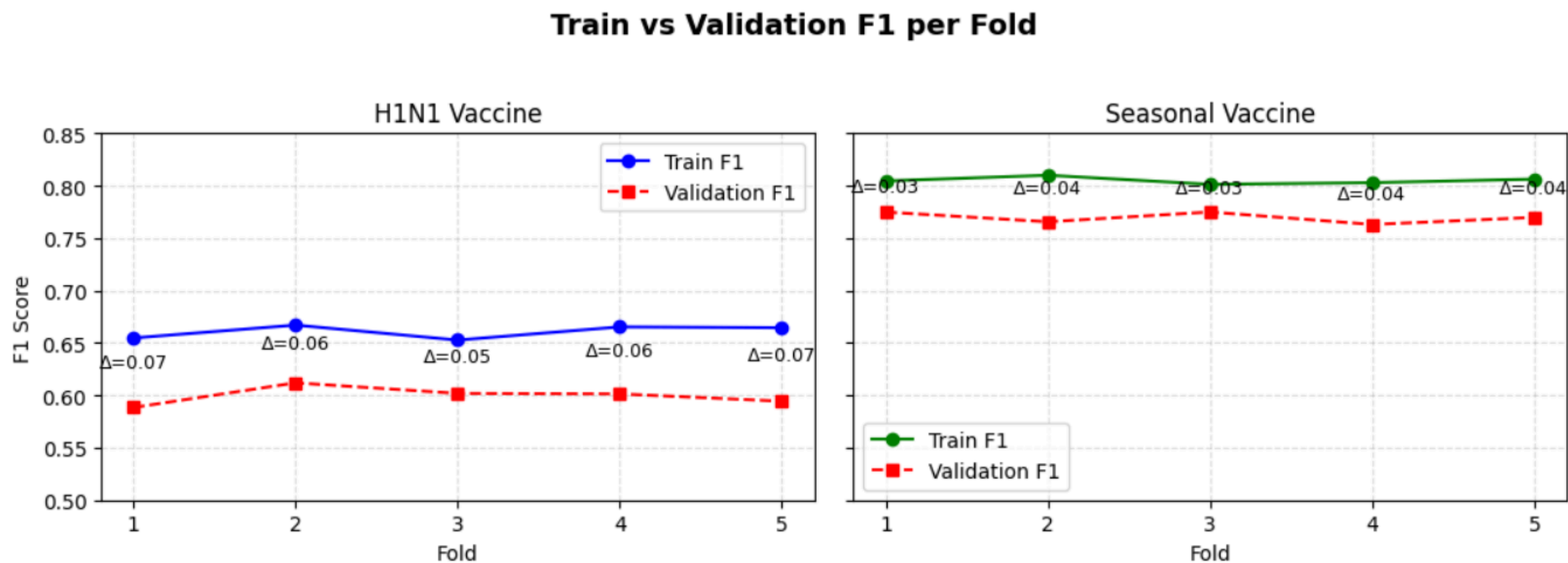
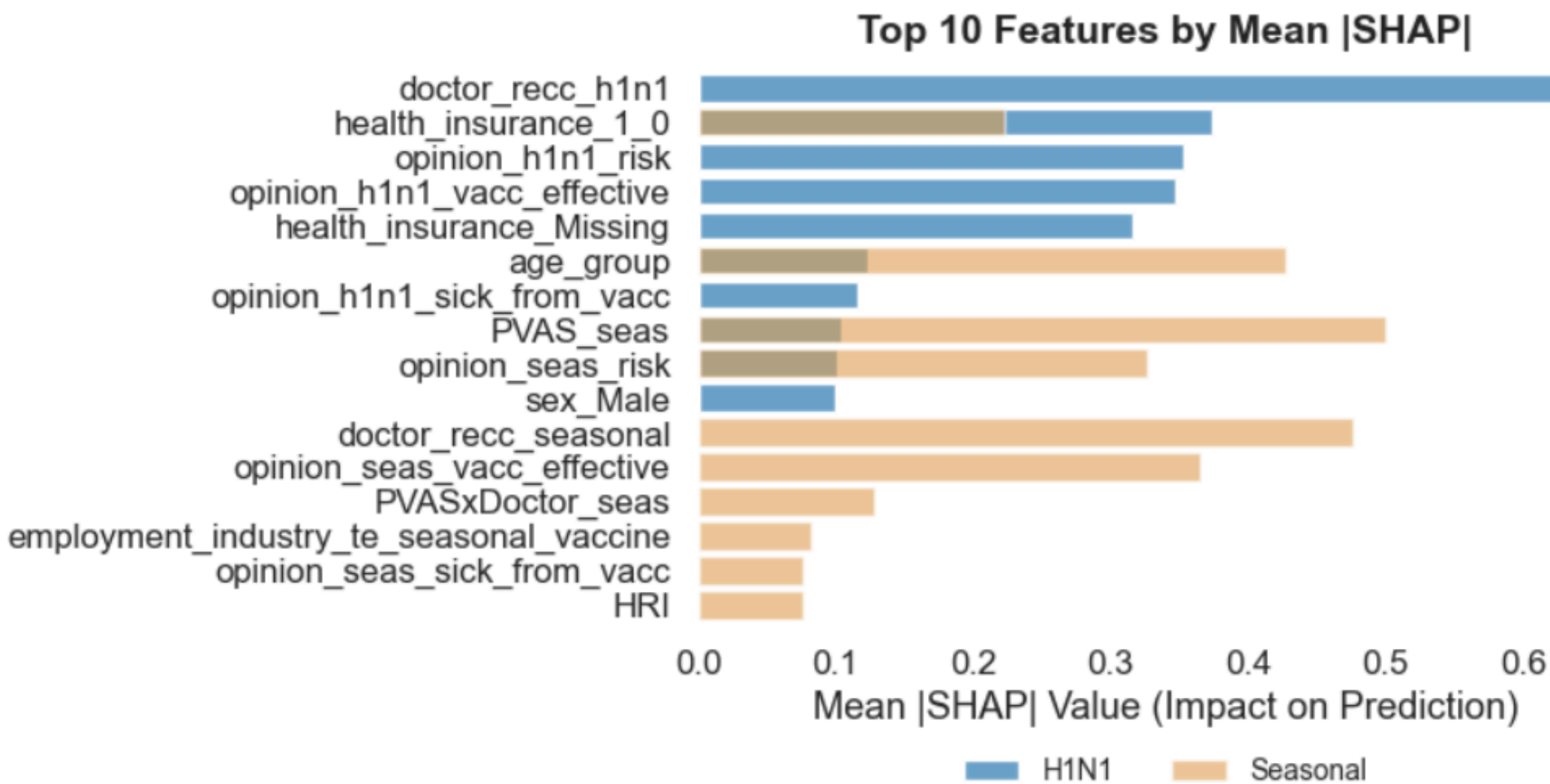
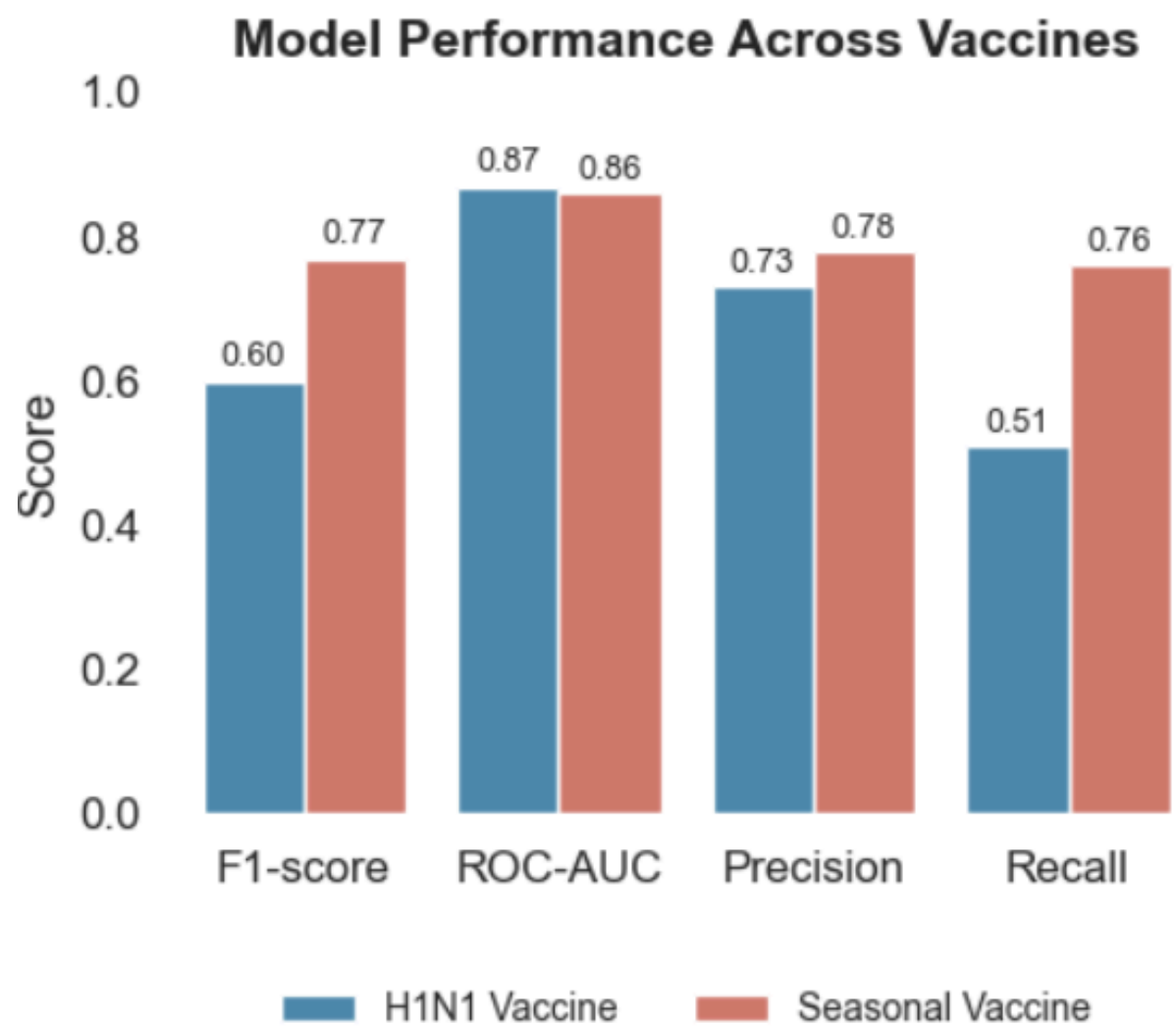
🏁 Final Decision

Select **LightGBM** as the final model for further tuning and SHAP analysis.

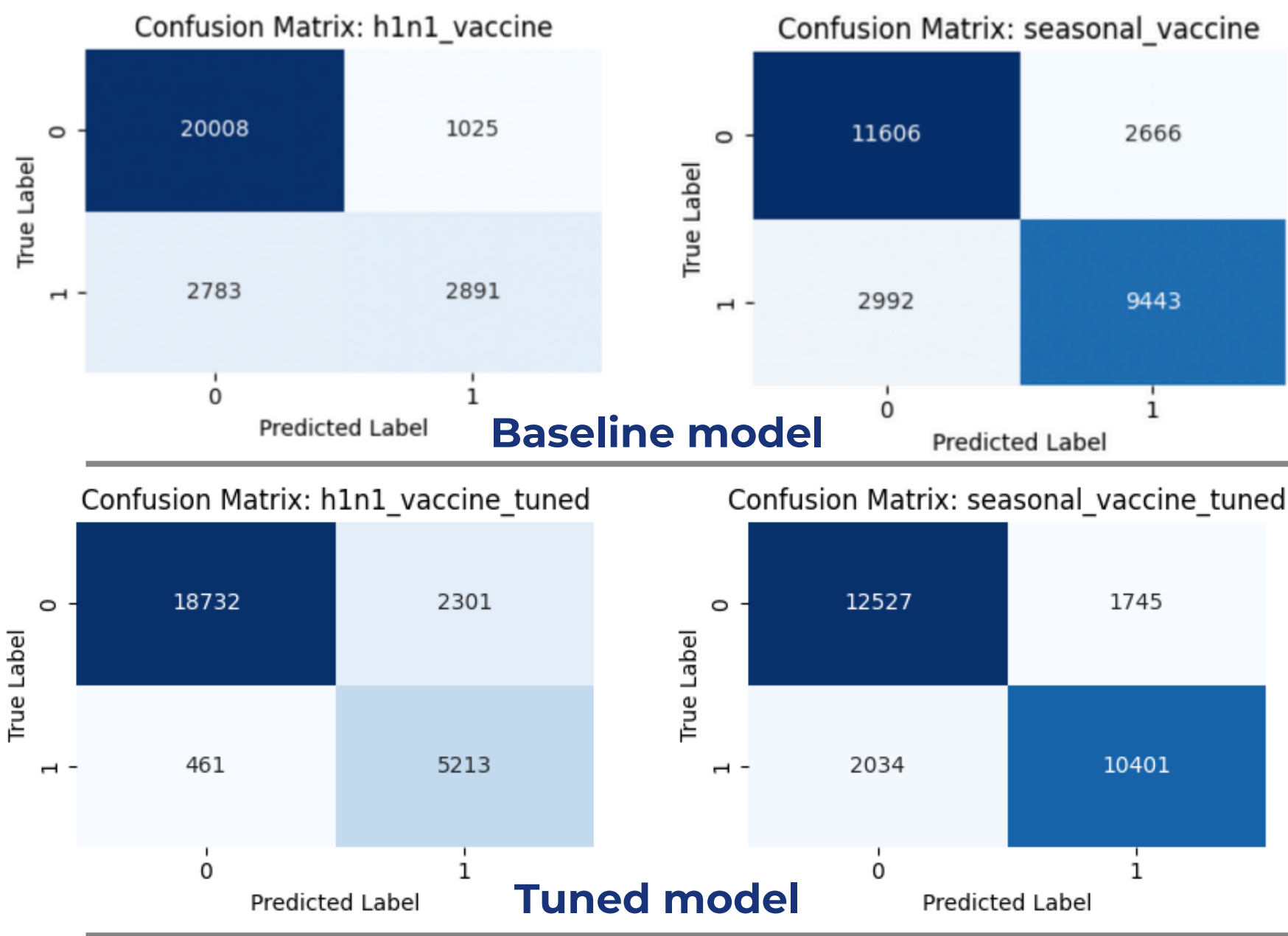
Why:

- It has the highest macro AUC.
- Offers the best balance between recall and precision across both vaccine labels.
- Supports advanced interpretability (SHAP) and hyperparameter fine-tuning.
- Scales efficiently to larger data and supports `scale_pos_weight` for H1N1 imbalance correction.

Baseline LightGBM Model & SHAP Analysis



Model Tuning with Class Imbalance Adjustment



- 1. **H1N1 model** — major improvement in recall and overall discrimination
 - The tuned model detects nearly twice as many vaccinated individuals as before (recall ↑ from 0.51 to 0.92)
 - Slight loss in precision (fewer overconfident false positives) is acceptable because the goal was to reduce false negatives — crucial when identifying under-vaccinated subpopulations for outreach
 - ROC-AUC = 0.97 confirms the model now separates classes much more effectively
 - The use of scale_pos_weight correctly re-balanced the learning focus toward the minority (vaccinated) class
- 2. **Seasonal model** — smaller but consistent gains
 - Both precision and recall improved simultaneously, indicating more stable and generalizable decision boundaries
 - ROC-AUC = 0.94 shows stronger overall ranking quality
 - Improvements came mostly from deeper trees and better feature sampling rather than class weighting

=== Baseline vs Tuned ===

Target	Baseline_F1	Tuned_F1	Baseline_Precision	Tuned_Precision	Baseline_Recall	Tuned_Recall	Baseline_ROC_AUC	Tuned_ROC_AUC	ΔF1
h1n1_vaccine	0.603	0.791	0.738	0.694	0.510	0.919	0.870	0.966	0.188
seasonal_vaccine	0.769	0.846	0.780	0.856	0.759	0.836	0.864	0.937	0.077

Sprint 2 Summary & Next Steps

What We Achieved

- Built and validated LightGBM models for H1N1 and seasonal flu vaccination.
- Applied domain-driven feature engineering to capture behavior, attitude, and health risk.
- Performed model benchmarking → LightGBM achieved best ROC-AUC.
- Fine-tuned models for F1 optimization to improve real-world prediction balance.
- Used SHAP analysis to explain key drivers of vaccination — trust, access, and risk.

Key Insights

- Doctor recommendation and vaccine attitude are the strongest predictors of uptake.
- Healthcare access and age shape seasonal vaccination patterns.
- Models are accurate, interpretable, and actionable for public-health planning.

Next Steps

- Train final models on full data with best-tuned parameters.
- Evaluate feature reduction — test smaller, high-impact subsets.
- Develop a “light” deployable version balancing speed and accuracy.
- Prepare explainability visuals (SHAP) for stakeholder reporting and deployment.

+ Thank
You For Your Attention



https://github.com/americano-diana/neuro_flushot/

